

Phase 9 - README

- [PHASE 9 — Multi-Agent Decision Intelligence](#)
 - [Real Estate AI Decision Intelligence Platform](#)
 - [What was built](#)
 - [Notable findings](#)
 - [How to run](#)
 - [Example output](#)

PHASE 9 — Multi-Agent Decision Intelligence

Real Estate AI Decision Intelligence Platform

The integration phase: Phases 4-8 are wired into one controlled decision workflow that answers “*Should I invest in Property X?*” end-to-end. Matches the established convention: `src/<topic>/*.py` modules (all Phase 4-8 engines reused **unchanged**), orchestrators, numbered CSVs, charts, automated tests.

What was built

11 agents in `src/agents/`, each returning a standard `AgentFinding`:

Agent	Reuses (unchanged)
<code>query_understanding.py</code>	— (new, deterministic parsing)
<code>data_agent.py</code>	Phase 1 tables directly
<code>market_agent.py</code>	Phase 8’s <code>_market_intelligence</code> + <code>_market_strength_score</code>
<code>prediction_agent.py</code>	Phase 4’s <code>build_property_master</code> + trained models, respects READY/CONDITIONAL/REJECTED
<code>evidence_agent.py</code>	Phase 5’s <code>HybridRetriever</code> + <code>rerank</code>
<code>finance_agent.py</code>	Phase 6’s <code>run_financial_engine</code> + <code>run_scenarios</code>
<code>risk_agent.py</code>	Phase 7’s <code>run_monte_carlo</code> + <code>compute_risk_metrics</code> + <code>compute_decision_stability</code>
<code>geo_agent.py</code>	Phase 8’s <code>run_geo_engine</code>
<code>evidence_validator.py</code>	Phase 5’s <code>grounding.validate_answer</code>
<code>decision_policy.py</code>	— (new, deterministic Decision Policy)
<code>explanation_agent.py</code>	— (new, deterministic template)

`orchestrator.py` wires all of them into the full §9.9 flow and assembles the final `DecisionResult`.
`test_phase9.py` — 37 automated tests, all passing (includes 1 post-delivery audit-fix regression test).

Full detail: `docs/phase9_report.md`, `docs/limitations.md`.

Notable findings

Two real integration bugs were found and fixed while building this phase — both documented in detail in the report: 1. **Module name collisions** — Phases 6/7/8 each had a `data_loader.py`; combined in one process, Python silently returned the wrong one. Fixed with two filename-only renames. 2. **Evidence citation cross-contamination** — a text-joining bug caused citation markers to attach to the wrong document’s numbers, which the Evidence Validator correctly caught as `BLOCKED`. Fixed, with a regression

test locking it in.

All four decision states (INVEST, HOLD, AVOID, INSUFFICIENT) are confirmed reachable with real properties. Decision confidence is permanently capped below ~82/100 because Phase 7's risk simulation is permanently **UNCALIBRATED** — this is reported honestly, not hidden.

How to run

```
cd phase9
pip install -r requirements-phase9.txt --break-system-packages

# Single property, full pipeline
python3 -c "
from src.agents.orchestrator import run_decision_pipeline
out = run_decision_pipeline('Should I invest in property 2731?', source='csv')
print(out['explanation'])
"

# Demo batch -> CSVs + charts
python3 src/agents/batch_runner_decision.py --source csv --n 25 --sims 1000
python3 src/agents/generate_charts.py --source csv --n 25 --sims 1000

# Tests
python3 -m pytest src/agents/test_phase9.py -q
```

With a live database, drop `--source csv` (defaults to `--source db`) and set `DB_USER / DB_PASSWORD`.

Example output

```
DECISION: INVEST (property_id=2731)
Decision score: 69.2/100 | Decision confidence: 81.7/100

Key reasons:
- Base-case IRR 12.9%, NPV ₹2,847,332 (5-year hold).
- Locality 14: demand index 91.2, supply index 38.7, absorption rate 2.1 (as of 2025-12-01).
- Location score 58.3/100.
...
```